The Missing Rung: AI and the Entry-Level Tech Pipeline

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The Challenge

Generative AI is changing the economics of entry-level work. Tasks that once gave junior designers and engineers their first opportunity to learn—producing design assets, writing boilerplate code, and building first-draft interfaces—can increasingly be automated or completed with AI assistance. The result may be less visible in total employment than in the pipeline into experienced work: if firms can obtain junior-level output without hiring juniors, fewer workers may enter the profession and fewer may eventually become experienced workers.

The question is therefore not simply whether AI reduces demand for junior workers, but whether the market will correct the resulting training shortfall before missing junior cohorts become a shortage of experienced talent.

This paper examines four occupations with different expected AI exposure: Graphic Designers, Software Developers, Web & Digital Interface Designers, and Construction Laborers. Software Developers provide a benchmark (Brynjolfsson et al., 2026); Graphic Designers provide the primary comparison; Interface Designers provide a more augmentation-oriented contrast; and Construction Laborers provide a lower-exposure control. The analysis does not claim that AI caused all observed employment changes.

The Economics

Three economic concepts explain why the entry-level pipeline can weaken.

First, training creates a positive externality. A firm bears the cost of developing a junior worker but may not capture the full return if that worker later leaves for another employer. Firms therefore have an incentive to invest less in training than would be optimal for the industry as a whole.

Second, the problem reflects short-run versus long-run elasticity. Firms can reduce junior hiring quickly when technology changes, but the supply of experienced workers cannot

respond as quickly because experience takes time to develop. A firm can eliminate a junior position today but cannot create an experienced replacement tomorrow.

Third, the specificity rule suggests targeting the market failure rather than the technology. If the problem is underinvestment in training, the response should restore investment in early-career development rather than restrict AI adoption. If the market failure is underinvestment in training, the policy should target the training externality rather than restrict the technology creating the change.

Analysis

The first test asks whether the observed early-career employment pattern follows the predicted AI-exposure mechanism.

Using IPUMS Current Population Survey data (Flood et al., 2022), I compared employment among 22–25-year-olds in Graphic Designers and Web & Digital Interface Designers between a pooled baseline of 2022 Q1–2023 Q4 and a current period of 2024 Q4–2026 Q3. Graphic Designers declined 38.3%, while Interface Designers declined 51.3%. The difference (Graphic Designers minus Interface Designers) was −13.0 percentage points, with a 95% confidence interval of −38.6 to +12.6 points.

This does not support the original hypothesis that Graphic Designers would experience the larger decline: the confidence interval includes zero, the point estimate runs opposite the prediction, and the small Interface Designer sample makes the estimate imprecise. The result is therefore inconclusive rather than evidence against AI disruption generally.

The Construction Laborers control declined only 1.83%, consistent with the roughly flat pattern expected of a low-exposure occupation. The relatively flat result is consistent with the control behaving as expected and provides less evidence of a common decline across the occupations.

For comparison, the same-source CPS data show a 1.6% decline in employment among 22–25-year-old Software Developers in our calculation, substantially smaller than the primary

study's (Brynjolfsson et al., 2026) ADP-based estimate of a roughly 19% relative employment shortfall among young workers in the most AI-exposed occupations compared with less-exposed occupations. These estimates are not directly comparable because they come from different data sources and constructions, so this analysis does not attempt to reconcile them.

The second test asks a different question: what happens if reduced junior hiring persists, regardless of exactly how much of the initial decline is attributable to AI? A cohort model tracks junior hires into the future mid-level workforce. The primary case assumes three years for a junior worker to progress to mid-level, with five years as a sensitivity case. Under the Depressed scenario, junior hiring remains at its current reduced level through 2030; under Recovery, it returns to its pre-2022 trend by 2028.

The timing result supports the second test. Under Depressed, the modeled mid-level shortage begins in 2026 with a three-year transition and 2028 with a five-year transition. Because the model treats the three-year junior-to-mid transition as a fixed lag, reduced hiring in 2023 begins to affect the modeled mid-level stock in 2026. Under Recovery, the gap narrows but remains open through 2035.

These figures are scenario outputs, not forecasts: even if firms eventually restore junior hiring, new hires cannot immediately become the experienced workers earlier cohorts would have produced.

The first test does not establish that Graphic Designers experienced a larger AI-driven decline. The second test does not establish that Graphic Designers are experiencing the depressed scenario; it takes the broader early-career employment effect identified in the primary study (Brynjolfsson et al., 2026) as an input and asks what follows if that effect persists. The second test evaluates pipeline timing, not the occupation-specific mechanism evaluated by the first.

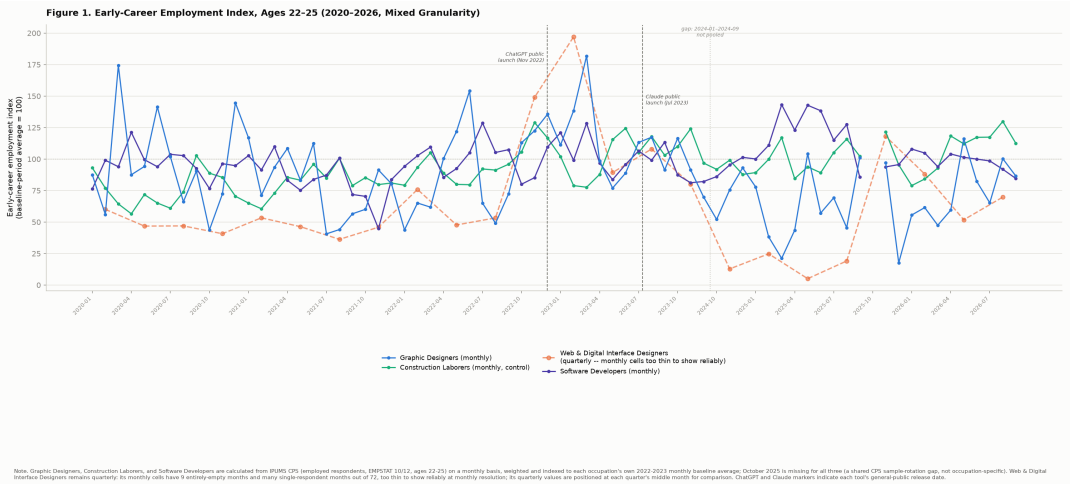


Figure 1. Early-career employment index, 2020–2026. Figure 1 compares early-career employment trajectories across the four occupations. It is descriptive rather than causal: it shows whether occupations experienced similar or different changes among workers ages 22–25 during the study period. The broader primary-study evidence addresses the distinction between early-career and experienced-worker outcomes. *Note.* All four series are calculated from IPUMS CPS data, weighted and indexed to each occupation's own baseline-period average. Graphic Designers, Construction Laborers, and Software Developers are shown at monthly resolution; Web & Digital Interface Designers remains quarterly, since its monthly samples are too thin to show reliably (many months have only one or two respondents, and several months have none).

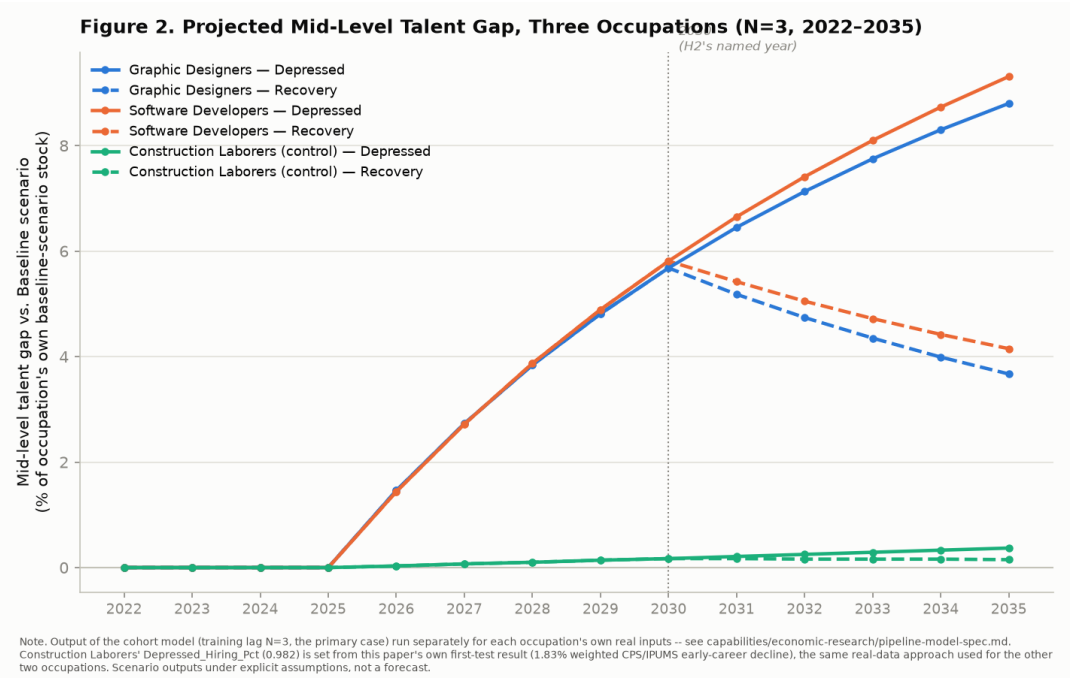


Figure 2. Projected mid-level talent gap, three occupations, 2022–2035. Output of the cohort model (training lag N=3, the primary case) run separately for each occupation's own real inputs. Gap is expressed as a percentage of each occupation's own baseline-scenario stock. Construction Laborers' depressed-hiring assumption (0.982) is based on the paper's observed 1.83% early-career employment decline — the same real-data approach used for the other two occupations.

Policy Recommendation

The case for intervention does not rest on the first test proving that AI caused the Graphic Designer decline. It rests on the broader evidence that AI-exposed occupations have experienced disproportionate early-career employment declines (Brynjolfsson et al., 2026) and on the economic logic that firms may underinvest in training when they cannot capture its full return. The recommendation is therefore conditional: where AI-driven reductions in junior work create a training shortfall, collective investment can address the market failure more directly than restricting AI adoption.

First, one possible response is a sector-wide training levy on firms within the affected sector. A levy could fund shared early-career training and reduce firms' incentive to free-ride on other employers' investments.

Second, policymakers should provide the legal foundation and oversight, while firms and the sector body determine training priorities and delivery.

Third, complement the levy with targeted apprenticeship funding or entry-level hiring credits to reduce the immediate cost and risk of hiring juniors.

Fourth, firms should redesign junior roles around AI oversight, judgment, and development, giving early-career workers responsibility for directing and correcting AI-generated work.

The strongest objection is that the market may eventually self-correct—or that the observed decline may not be sufficiently AI-specific to justify intervention. The evidence here cannot rule out the latter. But where AI has genuinely removed training opportunities, waiting for a mid-level shortage to trigger new hiring creates a timing problem: newly hired juniors still require years to become experienced workers.

The goal, therefore, is not to stop the market from adopting AI. It is to preserve the training function of the first rung while allowing the work on that rung to change.

References

Brynjolfsson, E., Chandar, B., & Chen, R. (2026). *Canaries in the coal mine? Six facts about the recent employment effects of artificial intelligence* (Rev. ed.). Stanford Digital Economy Lab.

Flood, S., King, M., Rodgers, R., Ruggles, S., Warren, J. R., & Westberry, M. (2022). *Integrated Public Use Microdata Series, Current Population Survey: Version 10.0* [Data set]. IPUMS. https://doi.org/10.18128/D030.V10.0

U.S. Bureau of Labor Statistics. (2025). *Occupational Outlook Handbook: Graphic designers*. U.S. Department of Labor.

U.S. Bureau of Labor Statistics. (2025). *Occupational Outlook Handbook: Web developers and digital designers*. U.S. Department of Labor.

U.S. Bureau of Labor Statistics. (2019). *Occupational Outlook Handbook: Software developers*. U.S. Department of Labor.

U.S. Bureau of Labor Statistics. (2025). *Occupational Outlook Handbook: Home health and personal care aides*. U.S. Department of Labor.